

Disproof of conjecture 00000001680:  
at the connectivity threshold the treewidth of a random geometric  
graph is at least  $c \log n$ , so it is not  $O(\sqrt{\log n / \log \log n})$

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## 1 The conjecture and the clause refuted

**Statement (English, verbatim).** Conjecture: The treewidth of random geometric graphs (unit disk, radius  $r_n$  at the connectivity threshold) is  $\Theta(\sqrt{\log n / \log \log n})$  (random geometric treewidth). The Chinese line says the same: the treewidth of random geometric graphs (unit disk, radius  $r_n$  at the connectivity threshold) is  $\Theta(\sqrt{\log n / \log \log n})$ .

**Reading.**  $\Theta(f)$  is the conjunction of a lower bound  $\Omega(f)$  and an upper bound  $O(f)$ , with  $f(n) = \sqrt{\log n / \log \log n}$ . We refute the upper bound:

$$(U) \quad \exists C \text{ such that } \text{tw}(G_n) \leq C f(n) \text{ for large } n$$

in each of the usual probabilistic senses (with probability tending to one, almost surely, or for the expectation). This refutes the conjunction. We do *not* address the lower bound  $\Omega(f)$  (it is true, and much weaker than what we prove). We prove more than needed: for every constant  $C$  and all large  $n$ , the bound  $\text{tw}(G_n) \leq C f(n)$  fails for *every* placement of the points, so the event in (U) is empty and has probability 0 under every distribution of the points.

**Reading choices covered.**

- *Region.* Any box  $[-L, L]^2$ ,  $L > 0$ . This contains the unit square  $[0, 1]^2$ , the unit disk, and every bounded region. The model on  $[0, \sqrt{n}]^2$  with radius  $\rho$  is the model on  $[0, 1]^2$  with radius  $\rho/\sqrt{n}$  (scale the points by  $1/\sqrt{n}$ ; the graph is unchanged), and  $n(\rho/\sqrt{n})^2 = \rho^2$ , so it is covered too.
- *Radius regime.* Any  $r > 0$  with  $nr^2 \geq c_0 \log n$ , for any fixed  $c_0 > 0$ . The connectivity threshold  $\pi nr_n^2 = \log n + c$  (or  $+c_n$  with  $c_n \geq -\frac{1}{2} \log n$ ), and  $\pi nr_n^2 \sim \log n$ , all satisfy this for large  $n$  with  $c_0 = 1/(2\pi)$ .
- *Point model.* The theorem is for  $m$  points with  $m \geq n/2$  and the radius condition stated in terms of  $n$ . So it covers the binomial model ( $m = n$ ) and the Poisson model of intensity  $n$  on the event  $\{N \geq n/2\}$ . On a region of area  $a$  the Poisson count has mean  $an$ , so this event has probability tending to one when  $a > 1/2$ , which holds for the unit square and the unit disk. For a smaller region, normalize  $n$  to the expected point count (or lower the count threshold and adjust the constants); the deterministic theorem is unchanged.
- *Metric.* Euclidean distance (as in the conjecture's "unit disk"). The Lean theorem holds for every supergraph of the Euclidean unit-disk graph on the same points; for example, on

$[0, 1]^2$  the flat-torus distance is at most the Euclidean distance, so the toroidal graph is such a supergraph.

## 2 Definitions (as formalized)

**Definition 1** (Unit-disk graph). For points  $p_1, \dots, p_m \in \mathbb{R}^2$  (Mathlib `EuclideanSpace` `ℝ` (Fin 2), Euclidean distance) and  $r > 0$ ,  $G(p, r)$  has vertex set  $\{1, \dots, m\}$ , and distinct  $i, j$  are adjacent iff  $|p_i - p_j| \leq r$  (`unitDiskGraph`).

**Definition 2** (Tree decomposition, treewidth). A tree decomposition of a graph  $G = (V, E)$  is a finite tree  $T$  with a bag  $B_t \subseteq V$  at each node  $t$  such that (T1) every vertex lies in some bag; (T2) for every edge  $uv$  some bag contains both  $u$  and  $v$ ; (T3) for every vertex  $v$ , the nodes  $t$  with  $v \in B_t$  induce a connected subgraph of  $T$ . Its width is  $\max_t |B_t| - 1$ . The treewidth  $\text{tw}(G)$  is the minimum width over all tree decompositions (`TreeDecomposition`, `width`, `treewidth`; the tree is a Mathlib `SimpleGraph.IsTree` on a finite node type, and (T3) uses Mathlib's `induce` and `Connected`).

## 3 Proof

**Lemma 3** (Helly property of subtrees; `helly_three`, `helly`). *In a tree, any finite family of node sets that each induce a connected subgraph and that pairwise intersect has a common node.*

*Proof.* Call  $S$  convex if any two nodes of  $S$  are joined by a walk inside  $S$ ; (T3) gives this. In an acyclic graph the path between two nodes is unique, so it lies in every convex set containing both ends (`path_subset`), and the intersection of two convex sets is convex (`wconn_inter`).

Three sets  $X, Y, Z$ : take  $a \in X \cap Y$ ,  $b \in Y \cap Z$ ,  $c \in Z \cap X$ . The path  $P$  from  $b$  to  $a$  lies in  $Y$ , and also in  $Z \cup X$  (it is the bypass of the walk  $b \rightarrow c \rightarrow a$  through  $Z$  then  $X$ ). Walking along  $P$  from  $b \in Z$  to  $a \in X$ , there are nodes  $x \in Z$ ,  $z \in X$  of  $P$  with  $x = z$  or  $xz$  an edge (`transition`). If  $x = z$  it is a common node. Otherwise, for the paths from  $c$  to  $x$  (inside  $Z$ ) and from  $c$  to  $z$  (inside  $X$ ): either the first passes through  $z$  or the second passes through  $x$  (if  $z$  is not on the first, appending the edge  $xz$  gives a path, which is the second by uniqueness; `edge_side`). So  $z \in Z$  or  $x \in X$ , and that node lies in  $X \cap Y \cap Z$ .

Finite families: induction on the family, carrying a convex set  $R$  that meets every pairwise intersection; adding a set  $S_a$  replaces  $R$  by  $S_a \cap R$ , and the three-set case shows the pairwise intersections still meet it.  $\square$

**Lemma 4** (`clique_subset_bag`, `card_clique_le_treewidth_add_one`). *If  $K$  is a clique of  $G$ , every tree decomposition has a bag containing  $K$ . Hence  $|K| \leq \text{tw}(G) + 1$ .*

*Proof.* Apply the Helly lemma to the sets  $\{t : v \in B_t\}$ ,  $v \in K$ : they are connected by (T3) and pairwise intersect by (T1) and (T2). The minimum in  $\text{tw}$  is attained (the one-bag decomposition shows the set of widths is nonempty).  $\square$

**Theorem 5** (Grid pigeonhole; `points_le_treewidth`, `log_le_treewidth`). *Let  $p_1, \dots, p_m \in [-L, L]^2$ ,  $r > 0$ , and  $G \supseteq G(p, r)$ . Then  $m \leq (\text{tw}(G) + 1)(32L^2/r^2 + 2)$ . If moreover  $n \geq 1$ ,  $n \leq 2m$  and  $c_0 \log n \leq nr^2$ , then  $\log n \leq A(\text{tw}(G) + 1)$  with  $A = 64L^2/c_0 + 4$ .*

*Proof.* Put  $s = r/2$ ,  $M = \lfloor 2L/s \rfloor$ , and send  $p_i$  to the cell  $(\lfloor (x_i + L)/s \rfloor, \lfloor (y_i + L)/s \rfloor) \in \{0, \dots, M\}^2$ . By pigeonhole some cell holds a set  $K$  of at least  $m/(M+1)^2$  points. Two points in one cell differ by less than  $s$  in each coordinate, so their distance is less than  $\sqrt{2}s < r$ :  $K$  is a clique, and

$|K| \leq \text{tw}(G) + 1$ . Since  $M \leq 4L/r$ ,  $(M+1)^2 \leq 2((4L/r)^2 + 1) = 32L^2/r^2 + 2$ . For the second claim use  $\log n/r^2 \leq n/c_0$  and  $\log n \leq n$ :  $n \log n \leq 2m \log n \leq 2(\text{tw}+1)(32L^2 \log n/r^2 + 2 \log n) \leq n A(\text{tw}+1)$ .  $\square$

**Theorem 6** (Main; `treewidth_not_upper_bound_supergraph`, `treewidth_not_upper_bound`). For all  $L > 0$ ,  $c_0 > 0$  and every real  $C$  there is  $N$  such that for all  $n \geq N$ , all  $m \geq n/2$ , all  $p_1, \dots, p_m \in [-L, L]^2$ , all  $r > 0$  with  $c_0 \log n \leq nr^2$ , and every supergraph  $G$  of  $G(p, r)$  (in particular  $G = G(p, r)$ ):

$$C \sqrt{\log n / \log \log n} < \text{tw}(G).$$

*Proof.* Let  $X = A|C| + A + 1$  and take  $N \geq e^{\max(X^2, e)}$ . For  $n \geq N$ ,  $\ell = \log n \geq \max(X^2, e)$ , so  $\log \ell \geq 1$  and  $x = \sqrt{\ell} \geq X \geq 1$ . Then  $C \sqrt{\ell / \log \ell} \leq |C| x$  and  $x^2 \geq Xx = A|C|x + (A+1)x > A|C|x + A$ , so  $|C|x < \ell/A - 1 \leq \text{tw}(G)$ .  $\square$

**Corollary 7** (`prob_upper_bound_eq_zero`). For  $n \geq N$  and  $r$  as above, every measure  $\mu$  on placements  $(\mathbb{R}^2)^n$  with  $\mu(\text{some point outside } [-L, L]^2) = 0$  gives  $\mu\{\text{tw}(G(p, r)) \leq C \sqrt{\log n / \log \log n}\} = 0$ . In particular this holds for  $n$  i.i.d. uniform points in the unit square or unit disk, for every  $C$ ; so (U) fails with probability one, and also fails for  $\mathbb{E} \text{tw}$  (since  $\text{tw} \geq \log n/A - 1$  pointwise).

*Remark 8* (Literature context). Mitsche and Perarnau, *On the treewidth and related parameters of random geometric graphs* (STACS 2012), abstract at <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.STACS.2012.408>: “We give asymptotically exact values for the treewidth  $\text{tw}(G)$  of a random geometric graph  $G(n, r)$  in  $[0, \sqrt{n}]^2$ . More precisely, we show that there exists some  $c_1 > 0$ , such that for any constant  $0 < r < c_1$ ,  $\text{tw}(G) = \Theta(\log(n)/\log \log(n))$ , and also, there exists some  $c_2 > c_1$ , such that for any  $r = r(n) > c_2$ ,  $\text{tw}(G) = \Theta(r \sqrt{n})$ .” Since  $\text{tw} \leq n - 1$ , the second formula can only hold in the unsaturated range (e.g.  $r = O(\sqrt{n})$ ; in general the expression saturates at  $\min(n, r\sqrt{n})$ ). At the connectivity threshold  $r \asymp \sqrt{\log n}$  in  $[0, \sqrt{n}]^2$ , which lies in that range, it gives  $\Theta(\sqrt{n \log n})$ . Our proof does not use this result.

## 4 Lean formalization

File `lean/Conjecture1680/Basic.lean` (Lean 4, Mathlib v4.33.1, only import `Mathlib`), namespace `C1680`. Main theorems: `treewidth_not_upper_bound`, `treewidth_not_upper_bound_supergraph`, `prob_upper_bound_eq_zero`; key lemmas: `helly`, `clique_subset_bag`, `card_clique_le_treewidth_add_one`, `points_le_treewidth`, `log_le_treewidth`. `lake build` succeeds, and `#print axioms` for each main theorem gives exactly `[propxt, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide`, and no added axiom.

Conventions: `Real.log` is the natural logarithm with  $\log 0 = 0$ , and  $\sqrt{\cdot}$  is `Real.sqrt` (0 on negatives); neither convention matters for  $n \geq N$ , where  $\log n \geq e$  and  $\log \log n \geq 1$ . Treewidth is a natural number; for the empty graph the one-bag decomposition has width 0 (truncated subtraction), which does not arise here since  $m \geq 1$ .